Aquablend — MILP Mathematical Formulation

Sprint 1 Network formulation, applied to the toy model

MILP Stream

July 25, 2026

1 Overview

This document presents the mathematical formulation of the Aquablend model: a single-period Mixed-Integer Linear Program (MILP) that chooses which water sources and treatment plants to operate, how much to draw from each source, and how to route water through the source → plant → zone network, so that demand and water-quality limits are met at least cost. It sets out the notation (sets, parameters, and decision variables), the objective, and the full set of constraints, then shows how the Sprint 1 toy model arises as the single-plant, single-zone case of the same formulation.

Convention. This is a single-period model (one representative day), so no day index is used. Parameters are uppercase; decision variables are lowercase, using Greek letters for binary decisions and lowercase Latin for continuous volumes. A subscript is an index; an overbar/underbar is an upper/lower bound.

2 Notation

2.1 Sets

Notation	Description
$\mathcal{S}$	set of water sources, $s \in \mathcal{S}$
$\mathcal{T}$	set of treatment plants, $t \in \mathcal{T}$
$\mathcal{Z}$	set of demand zones, $z \in \mathcal{Z}$
$\mathcal{P}$	set of water quality parameters, $p \in \mathcal{P}$

2.2 Parameters

Notation	Description	Units
D_z	volume of water to deliver to demand zone z	ML/day
F_s	fixed cost of activating source s	\$/day
F_t	fixed cost of activating treatment plant t	\$/day
C_s	unit cost of drawing water from source s	\$/ML
C_t	unit treatment cost at plant t	\$/ML
$\overline{W}_s$	maximum daily withdrawal from source s	ML/day
$\overline{V}_t$	maximum daily throughput of plant t	ML/day
$\overline{L}_{st}$	maximum flow on the link from source s to plant t	ML/day
$\overline{L}_{tz}$	maximum flow on the link from plant t to zone z	ML/day
Q_{sp}	value of quality parameter p in source s	units of p
$\underline{Q}_p$	lower regulatory limit for parameter p	units of p
$\overline{Q}_p$	upper regulatory limit for parameter p	units of p

Costs and limits are treated as fixed. The units of a quality parameter depend on p (e.g. mg/L, NTU). Water quality parameters that are not linear in their natural units are stored in Q_{sp} as a transformed value that blends approximately linearly.

2.3 Decision variables

Notation	Domain	Description	Units
α_s	$\{0,1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day
β_t	$\{0,1\}$	1 if plant t is activated	—
γ_{st}	$\{0,1\}$	1 if source s sends water to plant t	—
b_{st}	≥ 0	volume of water from source s to plant t	ML/day
δ_{tz}	$\{0,1\}$	1 if plant t sends water to demand zone z	—
c_{tz}	≥ 0	volume of water from plant t to demand zone z	ML/day

3 Example network structure

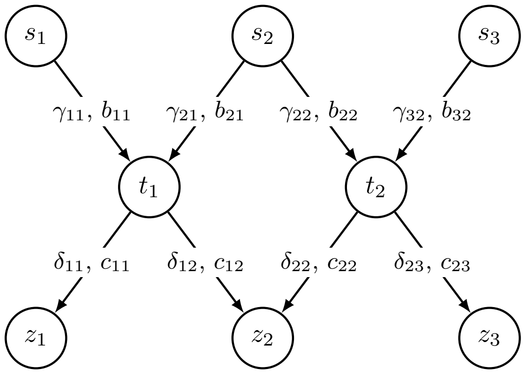

Figure 1: One example of the network structure (sources $\rightarrow$ treatment plants $\rightarrow$ demand zones). Each source-plant arc carries (γ_{st}, b_{st}) and each plant-zone arc carries (δ_{tz}, c_{tz}). The formulation is generic over any number of sources, plants, zones and the arcs connecting them; this is only one instance.

4 Objective

Minimise total cost over a single period: the fixed cost of activating each source and plant, the per-ML cost of drawing water, and the per-ML cost of treating it (charged on the volume each plant processes, $\sum_s b_{st}$).

$$\min \quad \underbrace{\sum_{s\in\mathcal{S}} F_s\alpha_s}_{\text{source activation}} + \underbrace{\sum_{t\in\mathcal{T}} F_t\beta_t}_{\text{plant activation}} + \underbrace{\sum_{s\in\mathcal{S}} C_s a_s}_{\text{drawing water}} + \underbrace{\sum_{t\in\mathcal{T}} C_t \sum_{s\in\mathcal{S}} b_{st}}_{\text{plant treatment}} \tag{1}$$

5 Constraints

Non-negativity. All continuous flow variables must be non-negative.

$$a_s, b_{st}, c_{tz} \geq 0, \qquad \forall s \in \mathcal{S}, t \in \mathcal{T}, z \in \mathcal{Z}. \tag{2}$$

Demand satisfaction. The total volume of water sent from all treatment plants to demand zone z via the links c_{tz} must be at least the demand D_z for that zone.

$$\sum_{t\in\mathcal{T}} c_{tz} \geq D_z, \qquad \forall z \in \mathcal{Z}. \tag{3}$$

Source capacity and activation. Water can only be drawn from a source if it has been activated, and never more than its maximum daily withdrawal. If the source is inactive ($\alpha_s = 0$) the draw a_s is forced to zero. If it is active ($\alpha_s = 1$), the draw is capped at $\overline{W}_s$.

$$a_s \leq \overline{W}_s\,\alpha_s, \quad \forall s \in \mathcal{S}. \tag{4}$$

Plant capacity and activation. A plant can process water only if it has been activated, up to its throughput capacity. The total volume arriving from all sources, $\sum_s b_{st}$, is capped at $\overline{V}_t$ and forced to zero when the plant is inactive ($\beta_t = 0$).

$$\sum_{s \in \mathcal{S}} b_{st} \le \overline{V}_t \beta_t, \quad \forall t \in \mathcal{T}. \tag{5}$$

Plant flow conservation. The total volume leaving plant t toward all zones via the links c_{tz} equals the total volume entering it from all sources via the links b_{st}.

$$\sum_{z \in \mathcal{Z}} c_{tz} = \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T}. \tag{6}$$

Source flow conservation. The volume drawn from each source a_s must equal the total volume it sends to all plants via the links b_{st}.

$$a_s = \sum_{t \in \mathcal{T}} b_{st}, \quad \forall s \in \mathcal{S}. \tag{7}$$

Link capacity. Each link has a maximum flow and can carry water only if that link is active. For a source-to-plant link, the flow b_{st} is capped at $\overline{L}_{st}$ and forced to zero unless the link is active ($\gamma_{st} = 1$).

$$b_{st} \le \overline{L}_{st} \gamma_{st}, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{8}$$

Similarly, a plant-to-zone link's flow c_{tz} is capped at $\overline{L}_{tz}$ and is zero unless the link is active ($\delta_{tz} = 1$).

$$c_{tz} \le \overline{L}_{tz} \delta_{tz}, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{9}$$

Links require active nodes. Links can only be active if the corresponding upstream node (source or treatment plant) is active. This is enforced by ensuring that the binary variables representing the links are less than or equal to the binary variables representing the activation of the sources and plants. A source-to-plant link requires the source to be active:

$$\gamma_{st} \le \alpha_s, \quad \forall s \in \mathcal{S}, t \in \mathcal{T}. \tag{10}$$

A plant-to-zone link requires the plant to be active:

$$\delta_{tz} \le \beta_t, \quad \forall t \in \mathcal{T}, z \in \mathcal{Z}. \tag{11}$$

Optional: a source-to-plant link could also require the plant to be active,

$$\gamma_{st} \le \beta_t, \quad \forall s \in \mathcal{S},\ t \in \mathcal{T}. \tag{12}$$

This is not explicitly needed, since zero flow to an inactive plant is already guaranteed by the plant capacity constraint (5). Suppose plant t is inactive, so $\beta_t = 0$. Then (5) becomes

$$\sum_{s \in \mathcal{S}} b_{st} \le \overline{V}_t \, \beta_t = 0,$$

and since every flow is non-negative ($b_{st} \ge 0$), this forces $b_{st} = 0$ for all $s \in \mathcal{S}$.

Water quality. The water arriving at each plant must stay within the quality limits. For parameter p, $\sum_s Q_{sp} b_{st}$ is the total amount of p entering plant t and $\sum_s b_{st}$ is the total volume, so this keeps the average value of p at the plant between $\underline{Q}_p$ and $\overline{Q}_p$.

$$\underline{Q}_p \sum_{s \in \mathcal{S}} b_{st} \le \sum_{s \in \mathcal{S}} Q_{sp} b_{st} \le \overline{Q}_p \sum_{s \in \mathcal{S}} b_{st}, \quad \forall t \in \mathcal{T},\ \forall p \in \mathcal{P}. \tag{13}$$

Each Q_{sp} must be a quantity that blends linearly by volume, so that summing each source's contribution, $\sum_{s \in \mathcal{S}} Q_{sp} \, b_{st}$, correctly represents the blended quality at the plant.

6 Application to the toy model

This network model is a generalisation of the Sprint 1 toy model, not a different problem: the toy is the special case with a *single* treatment plant and a *single* demand zone. With one plant and one zone there is nothing to route: every activated source sends its whole draw to the one plant ($b_{st} = a_s$) and all of it reaches the one zone ($c_{tz} = \sum_s a_s$), so the routing variables γ, δ, b, c are fully determined ($\gamma = \delta = 1$, $b_{st} = a_s$, $c_{tz} = \sum_s a_s$) and only the source decisions α_s, a_s remain free. Formally this is the network model with $\mathcal{T} = \{t\}$, $\mathcal{Z} = \{z\}$, the plant fixed active ($\beta_t = 1$) and the routing fixed on ($\gamma_{st} = \delta_{tz} = 1$). Substituting these into the network equations gives the toy model formulation below.

6.1 Decision variables

Notation	Domain	Description	Units
α_s	$\{0, 1\}$	1 if source s is activated	—
a_s	≥ 0	volume drawn from source s	ML/day

6.2 Objective

Minimise the fixed activation cost of the chosen sources plus the per-ML cost of drawing and treating the water. The single plant is always on, so its fixed cost F_t is a constant and is dropped.

$$\min \quad \underbrace{\sum_{s \in \mathcal{S}} F_s \alpha_s}_{\text{source activation}} + \underbrace{\sum_{s \in \mathcal{S}} C_s a_s}_{\text{drawing water}} + \underbrace{C_t \sum_{s \in \mathcal{S}} a_s}_{\text{plant treatment}} \tag{14}$$

6.3 Constraints

Non-negativity. The draw from each source must be non-negative.

$$a_s \geq 0, \qquad \forall s \in \mathcal{S}. \tag{15}$$

Demand satisfaction. The total volume drawn (all of which is delivered to the one zone) meets the zone's demand D_z.

$$\sum_{s \in \mathcal{S}} a_s \geq D_z. \tag{16}$$

Source capacity and activation. Identical to constraint (4).

Water quality. The blend of all drawn sources must lie within each parameter's limits.

$$\underline{Q}_p \sum_{s \in \mathcal{S}} a_s \leq \sum_{s \in \mathcal{S}} Q_{sp} a_s \leq \overline{Q}_p \sum_{s \in \mathcal{S}} a_s, \quad \forall p \in \mathcal{P}. \tag{17}$$

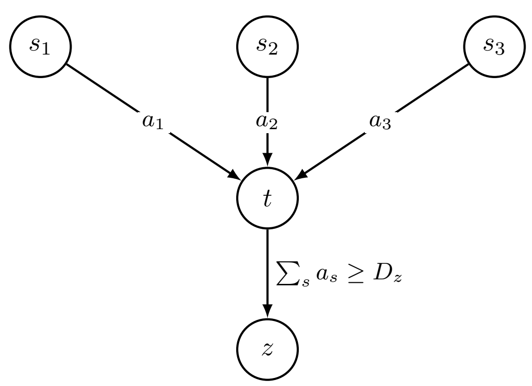

Figure 2: Toy model: three sources s_1, s_2, s_3 feed one plant t, which delivers to one zone z. Routing is fixed, so the delivered blend $\sum_s a_s$ must meet demand D_z and the quality limits.

Since this is just the network model with routing fixed, going the other way needs nothing new: free the routing variables and add more sources, plants and zones. The toy can therefore be built and tested first, then scaled up without reformulating.